



The husband-and-wife team Irène and Frédéric Joliot-Curie continued the work of Marie Curie—Irène's mother—after she died in 1934.

FRENCH CHEMISTS Irène (1897–1956) and Frédéric Joliot-Curie (1900–58) demonstrated that it was possible to induce **artificial radioactivity**. They created nuclear reactions by making high-energy particles collide with the nuclei of atoms. In February, they published their findings on making radioactive isotopes (variants) of phosphorus and nitrogen by firing alpha particles at nonradioactive targets.

Meanwhile, Italian physicist **Enrico Fermi** (1901–54) decided to use recently discovered subatomic particles called **neutrons** (see 1932–33), instead of alpha particles, in nuclear reactions. He worked his way up the periodic table to see how different

Pistol shrimp

The snapping sound of this shrimp's enlarged pincer produces enough energy in a fraction of a second to release heat and light as well as stun its prey.



elements captured the neutrons. At the time nobody thought that neutrons would have enough energy to split a heavier atom. But then, in January 1934, Fermi succeeded in splitting uranium by bombarding it with neutrons. Fermi thought he had made the first **transuranium element** (an element with an atomic number above 92, uranium), but he had in fact achieved **atomic fission** (see 1938).

German astronomer **Walter Baade** (1893–1960) and his Swiss counterpart **Fritz Zwicky** suggested that the smallest, densest stars that could be detected were made of neutrons. It was later confirmed that a **neutron star** is the remnant of a star that has exploded and is toward the end of its life.

While experimenting with ultrasound, German scientists **H. Frenzel** and **H. Schultes** noticed that high-frequency sound affected photographic

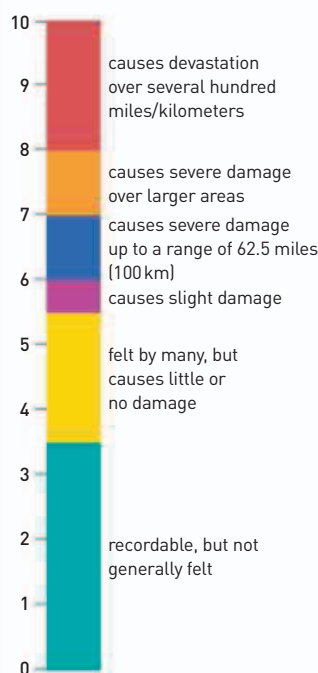
plates—indicating light production. Known as **sonoluminescence**, this occurs when sound creates underwater bubbles that focus its energy a trillion times, creating flashes of light and heat. In the natural world, pistol shrimps use this process to stun their prey.



In the rain forest's ecosystem, the living organisms—vegetation and animals—interact with the nonliving air and soil through the release and absorption of carbon dioxide and the exchange of nutrients.

INSTRUMENTS TO MEASURE

EARTHQUAKES date back to antiquity, but by the early 1900s seismometers were being used to detect movements in Earth's crust. A sophisticated system of levers generated a paper trace showing the magnitude of an earthquake. Working in earthquake-prone California, American physicist **Charles Richter** (1900–85) developed a



Richter scale

By linking measurable earthquake movements to environmental effects, Richter created a scale that could be understood by everybody.

480 MEGATONS

THE ENERGY RELEASED IN AN EARTHQUAKE MEASURING 9 ON THE RICHTER SCALE

scale that would make it easier to compare the amount of energy released by different earthquakes. Initially devised just for local use, the **Richter scale** was soon applied throughout the world.

In February, German bacteriologist **Gerhard Domagk** (1895–1964), described a chemical dye with powerful antibacterial properties. He reported on clinical trials suggesting that the dye—later sold under the name **Prontosil**—could be used as a drug to cure common, dangerous infections. Later in the year, Italian pharmacologist **Daniel Bovet** (1907–92) discovered the chemical basis for Prontosil's effectiveness. He found its active ingredient to be a sulfur-based compound called **sulfonamide**. Sulfonamides were the most important **antibacterial drugs** until the introduction of penicillin (see 1940–41).

During 1935, American chemist **Wallace Carothers** (1896–1937) led a team of scientists at

DuPont chemical laboratories in the development of new polymers—long molecules made by bonding smaller molecular building blocks into chains. The team had already manufactured an artificial silk called **polyester**, but was now working with a different kind of building block. The resulting polymer, called polyamide 6-6, could be drawn out into tough filaments. It was later called **nylon** (see 1937).

In July, a scientific paper in the journal *Ecology* introduced the concept of an **ecosystem**. Its author, British botanist **Arthur Tansley** (1871–1955), combined two key themes: American botanist Frederic Clements's 1916 concept of vegetation as a community of different species that changes over time, and Russian scientist Vladimir Vernadsky's treatise on cycles of matter (see 1926). Tansley's ecosystem was an ecological structure in which living organisms interacted with each other as well as with the

January Enrico Fermi unwittingly achieves **atomic fission**

February 10 Description of artificial induction of radioactivity is published

May It is suggested that transition of normal stars to **neutron stars**

August 15 American diver Otis Barton and naturalist Charles Beebe dive to a record 3,028 ft using a **bathysphere**

July 4 Hungarian physicist Leo Szilard files a patent for the means of producing a **nuclear chain reaction**—the nuclear bomb

H. Frenzel and H. Schultes describe **sonoluminescence**

January Charles Richter invents a scale for the magnitude of earthquakes

February 15 Gerhard Domagk describes the development of **Prontosil**—the first antibacterial drug

February 28 Wallace Carothers and his team produce polyamide 6-6, later called **nylon**

July Arthur Tansley introduces the concept of the **ecosystem**

August 16 Rudolf Schoenheimer describes the use of **isotopic labels** to analyze metabolism